

Student Number _____

ASCHAM SCHOOL



2020

YEAR 12

TRIAL

EXAMINATION

Mathematics

Extension 2

General Instructions

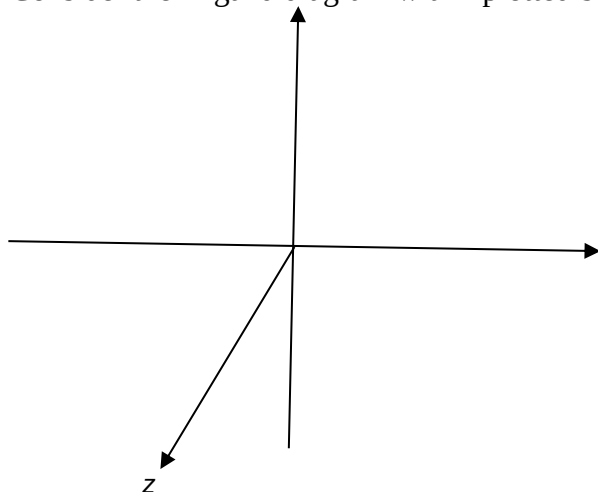
- Reading time – 10 minutes.
- Working time – 3 hours.
- Write using black non-erasable pen.
- NESA-approved calculators may be used.
- A NESA Reference Sheet is provided.
- All necessary working should be shown in every question.

Total marks – 100

- Attempt Sections A and B.
- Section A is worth 10 marks.
- Recommended time on Section A: 15 minutes
- Answer Section A on the multiple choice answer sheet.
- Detach the multiple choice answer sheet from the back of the examination paper.
- Section B contains 6 questions worth 15 marks each.
- Recommended time on Section B: 2 hours 45 minutes
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher.

SECTION A – 10 MULTIPLE CHOICE QUESTIONS 10 MARKS**ANSWER ON THE ANSWER SHEET**

- 1 Consider the Argand diagram with z plotted below.



Which of the following could be the complex number z ?

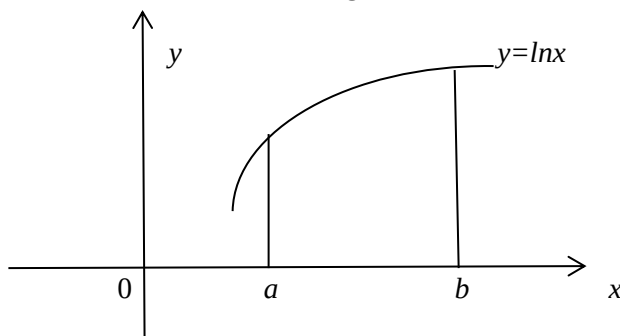
- A $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$
- B $\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right)$
- C $-\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}$
- D $\cos\left(-\frac{2\pi}{3}\right) - i \sin\left(-\frac{2\pi}{3}\right)$
- 2 Two of the roots of the equation $z^4 + Bz^3 + Cz^2 + Dz + E = 0$, where B, C, D, E are real are $3+i$ and $1-2i$. The value of E is:
- A -50
- B -24
- C 24
- D 50

- 3 Consider the proposition:

$$P(n): \sum_{k=1}^n (2k-1) = n^2.$$

Which of the following is true for $P(k+1)$?

- A $(2k-1) + (2k+1) = (k+1)^2$
 B $\sum_{1}^k (2k-1) + 2k = (k+1)^2$
 C $\sum_{1}^k (2k-1) + 2(k+1) = (k+1)^2$
 D $\sum_{1}^k (2k-1) + (2k+1) = (k+1)^2$
- 4 If $f(x) = \ln x$ is the continuous, strictly increasing function on the interval $[a, b]$, as shown below, which of the following three statements must be true?



I $\int_a^b \ln x dx < (b-a) \ln b$

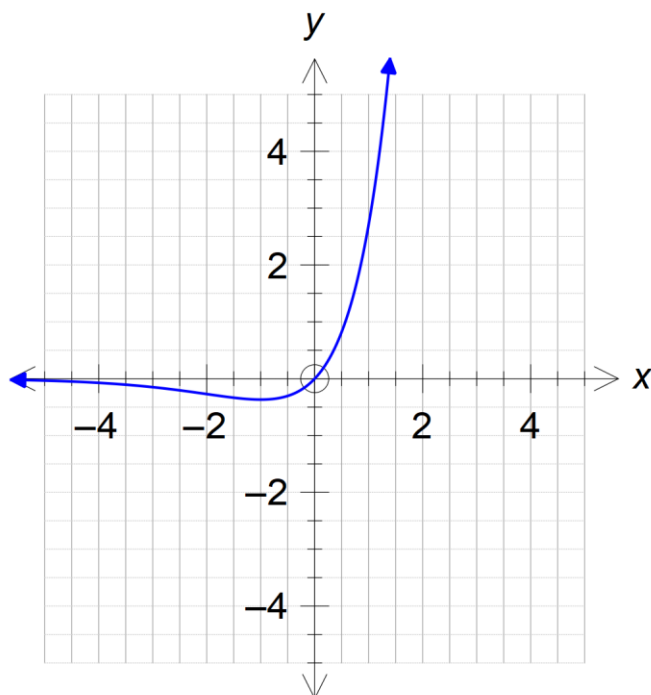
II $\int_a^b \ln x dx > (b-a) \ln a$

III there exists a number c where $a < c < b$, such that $\int_a^b \ln x dx = (b-a) \ln c$

- A I only
 B II only
 C III only
 D I, II and III

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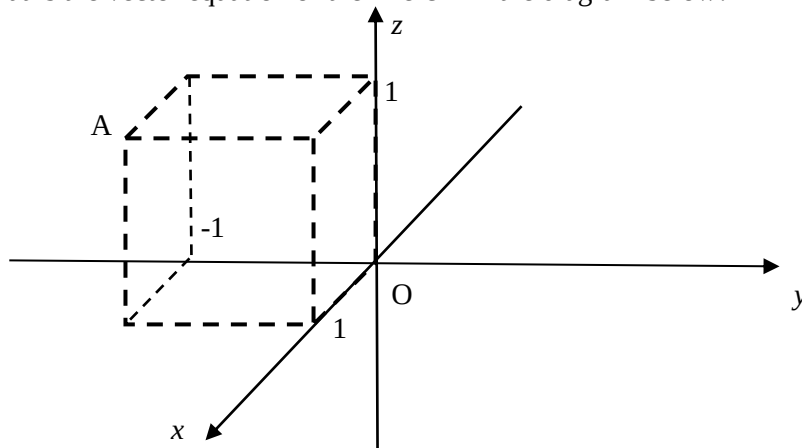
- 5 The velocity of a particle at $x = -3$ with simple harmonic motion described by $\ddot{x} = -12(x+3)$ where the amplitude is 4, is:
- A 1
- B 0
- C $\pm\sqrt{192}$
- D Insufficient information
- 6 Which of the following is a counter-example to the following statement?
All people who get ATARS over 99 do Extension 2 Mathematics.
- A No people who get ATARS over 99 do Extension 2 Mathematics.
- B Jesse did History and got an ATAR over 99.
- C Sam did Extension 2 Mathematics and got an ATAR under 99.
- D Polly did not do Extension 2 Mathematics and got an ATAR over 99.
- 7 The equation of the graph below could be:



- A $y = xe^x$
- B $y = xe^{-x}$
- C $y = x \ln x$
- D $y = -x \ln x$

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- 8 What is the vector equation of the line OA in the diagram below?



- A $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + k \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$
- B $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + k \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$
- C $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + k \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
- D $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + k \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

- 9 If $\frac{dQ}{dt} = k(Q - A)$ for $k, A \in \mathbb{R}$, then which of the equations below could be a solution?

- A $Q = k \left(\frac{Q^2}{2} - AQ \right)$ where $k, A \in \mathbb{R}$
- B $Q = Ae^{kt}$ where $k, A \in \mathbb{R}$
- C $Q = A + Be^{kt}$ where $k, A, B \in \mathbb{R}$
- D $Q = \frac{A}{1 + Be^{kt}}$ where $k, A, B \in \mathbb{R}$

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- 10** Consider the set of points described by $|z + 2| = 1$ on the complex plane. What is the maximum value of $\arg z$?

- A** $\frac{\pi}{6}$
- B** $\frac{\pi}{3}$
- C** $\frac{2\pi}{3}$
- D** $\frac{5\pi}{6}$

SECTION 2 – 6 QUESTIONS EACH WORTH 15 MARKS**Question 11 – Begin a new writing booklet**

a Find $\int \frac{x^3 dx}{x^8 + 3}$. 2

b Find $\int \cos^{-1} x \, dx$. 2

c Find $\int \sec^4 x \, dx$. 3

- d** The points A, B, C, D representing the complex numbers a, b, c, d form a parallelogram as shown in the diagram. $\angle ADC = \frac{2\pi}{3}$ and $|d - c| = |b - c|$.

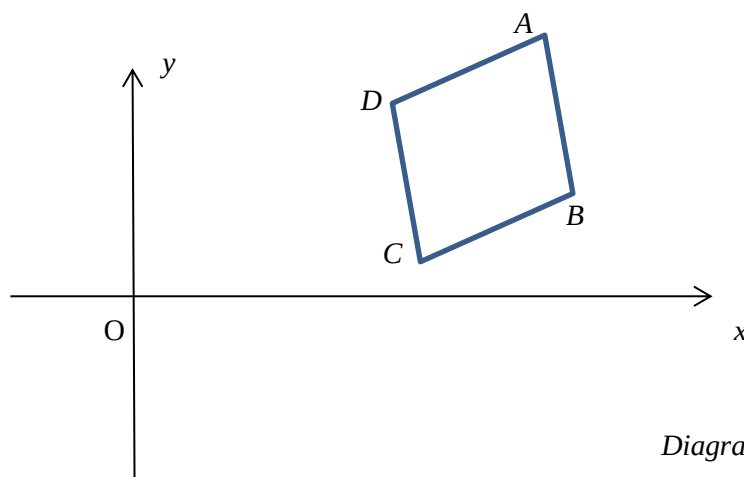


Diagram not to scale

Copy the diagram into your booklet.

i Show that $d - c = (b - c) \left(\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)$. 2

ii Find $\arg \left(\frac{a - d}{b - d} \right)$. 2

iii Find the value of $\left| \frac{b - a}{d - b} \right|$. 2

iv Explain why $\arg \left(\frac{a - c}{b - d} \right) = \frac{\pi}{2}$. 2

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Question 12 – Begin a new writing booklet

- a** Consider the statement for $n \in \mathbb{N}$:

If n is prime then n has exactly two factors.

- | | | |
|------------|---|---|
| i | Write the converse. | 2 |
| ii | Write the contrapositive. | 2 |
| iii | Determine whether or not the statement is an equivalence. Give reasons. | 2 |
| iv | Georg said that: | 2 |

If n does not have exactly two factors then n is composite.

Determine whether or not this statement is true. If not, give a counter-example.

- | | | |
|----------|--------------------------------------|---|
| b | Write the negation of the statement: | 2 |
|----------|--------------------------------------|---|

All meerkats eat grubs.

- | | | |
|----------|--|---|
| c | If $x, y \in \mathbb{R}$, solve for x and y : | 3 |
|----------|--|---|

$$x + 2y - 3xi + 4yi = 5 + 15i.$$

- | | | |
|----------|---|---|
| d | If $(x + iy)^6 = e^{2i\pi}$, where x, y are real, find a non-zero solution for x and y . | 2 |
|----------|---|---|

Question 13 – Begin a new writing booklet**a**

The two lines $\vec{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$ and $\vec{q} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ intersect at (a, b, c) .

i Find (a, b, c) . **2**

ii Show that $\vec{r}$ and $\vec{q}$ are perpendicular. **2**

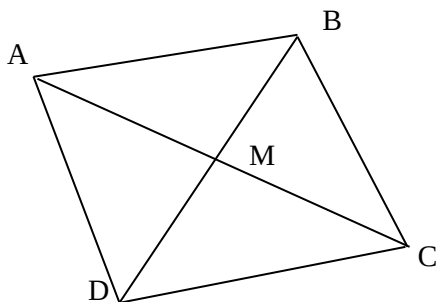
iii Find a vector which is perpendicular to both $\vec{r}$ and $\vec{q}$. **3**

b

The point $P(5, 7, 2)$ lies on the sphere $(x-1)^2 + (y+3)^2 + (z-3)^2 = k^2$. Find the value of k . **2**

c

Consider the quadrilateral $ABCD$ shown. The diagonals AC and BD bisect each other at M . **3**



Copy the diagram.

Use vectors to prove that $\overrightarrow{AB} = \overrightarrow{DC}$.

[Hint: for convenience, let $\overrightarrow{AM} = \vec{p}$ and $\overrightarrow{MB} = \vec{q}$.]

d

Find $\sqrt{15-8i}$. **3**

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Question 14 – Begin a new writing booklet

a i If a, b are real, show that $a^2 + b^2 \geq 2ab$. **1**

ii Hence show that if a, b, c are real then $a^2 + b^2 + c^2 \geq ab + bc + ca$. **2**

iii Hence show that $3(a^4 + b^4 + c^4) \geq (a^2 + b^2 + c^2)^2$. **2**

b Let $I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx, n \geq 0$.

i Prove that $\frac{1}{n!} = e(I_{n-1} - I_n)$. **2**

ii Hence evaluate I_4 . **2**

c Find the four 4th roots of -1 and show them on an Argand diagram. **3**

d Prove by mathematical induction that $\forall k \in \mathbb{N}$ and **odd** n : **3**

$$\sum_{k=1}^n (-1)^{k-1} k^3 = \frac{(2n-1)(n+1)^2}{4}.$$

[Hint: the identity below might be useful:

$$2k^3 + 15k^2 + 36k + 27 = (2k+3)(k+3)^2.]$$

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Question 15 – Begin a new writing booklet

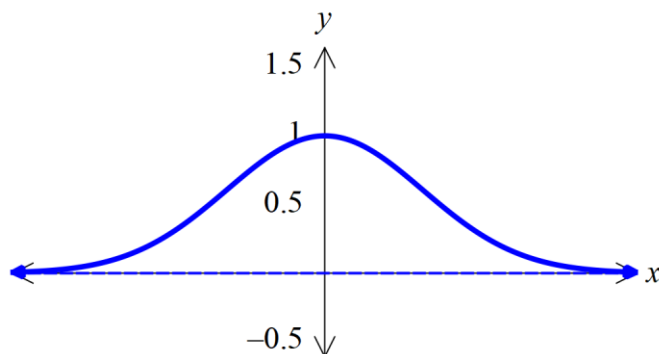
- a i** If $x + \frac{1}{x} = v$ find an expression for $x^3 + \frac{1}{x^3}$ in terms of v . 2

[Hint: expand $\left(x + \frac{1}{x}\right)^3$.]

- ii** Prove $x^5 + \frac{1}{x^5} = v^5 - 5v^3 + 5v$. 2

- iii** If $x = \cos \theta + i \sin \theta$, using the above parts, find $\cos 10\theta$ in terms of $\cos \theta$. 2

- b** For a certain function $f(x)$ the graph $y = f'(x)$ is sketched below.



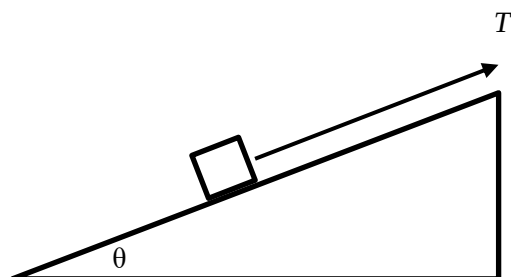
The graph $y = f(x)$ passes through $(0,0)$. The inverse of $f(x)$ is $g(x)$, that is $f^{-1}(x) = g(x)$.

- i** Sketch $y = f(x)$. 2
- ii** Sketch $y = \frac{d}{dx}(g(x))$. 2

Question 15 continues on the next page...

Question 15 continued...

- c A 2 kg mass is being pulled up a slope by a string of tension 10 Newtons. The slope is at an angle θ to the horizontal and the coefficient of friction is 0.3. As well as friction, the forces of gravity g and the normal are also acting. 2



Copy the diagram.

By resolving forces with components along and perpendicular to the slope, find the net force F_{net} in Newtons up the slope.

- d Use the substitution $u = \frac{1}{x}$ to evaluate $\int_1^{\infty} \frac{dx}{x\sqrt{x^2 + 2x - 1}}$. 3

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Question 16 – Begin a new writing booklet

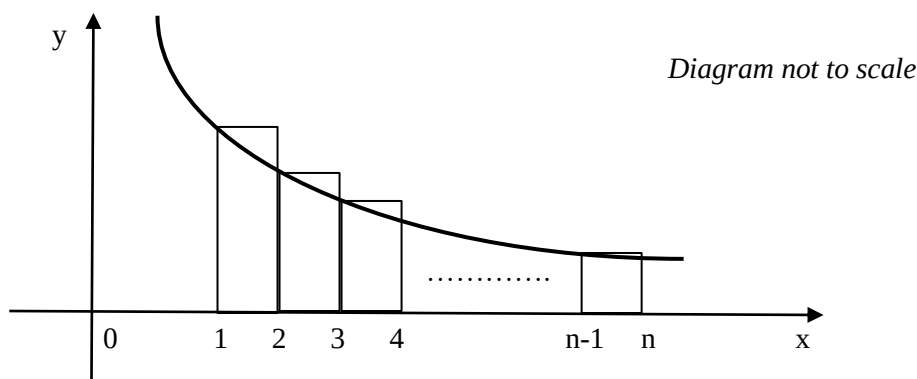
- a** The Covid-19 1.5 metre rule is based on the premise that a person emits droplets from a height 1.8 m above ground at an angle θ from the horizontal so that the maximum range on the ground is 1.5 metres. Show that the maximum speed V m/s of the droplets launched is given by **4**

$$V = \sqrt{\frac{15g}{22}}. \quad (\text{Assume there is no air resistance.})$$

[Hint: you can assume which angle gives the maximum range.....]

- b** Solve $\tan^{-1} 4x - \tan^{-1} 3x = \tan^{-1} \frac{1}{7}$. **2**

- c** Consider the curve $y = \frac{1}{x}$ sketched below with rectangles above the curve approximating the area under the curve between $x=1$ and $x=n$.



- i** Prove that $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n-1} > \ln n$. **4**
- ii** Does the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ have a limit? Give reasons. **1**
- d** Let ω be a complex cube root of unity. Prove that if $n \in \mathbb{N}$ then $1 + \omega^n + \omega^{2n} = 3$ if n is a multiple of 3 or 0 if n is not a multiple of 3. **4**

The end! ☺

Student Number

ASCHAM SCHOOL**YEAR 12 Trial Mathematics Extension 2 Exam****MULTIPLE-CHOICE ANSWER SHEET**1. A ☐ B ☐ C ☐ D ☐2. A ☐ B ☐ C ☐ D ☐3. A ☐ B ☐ C ☐ D ☐4. A ☐ B ☐ C ☐ D ☐5. A ☐ B ☐ C ☐ D ☐6. A ☐ B ☐ C ☐ D ☐7. A ☐ B ☐ C ☐ D ☐8. A ☐ B ☐ C ☐ D ☐9. A ☐ B ☐ C ☐ D ☐10. A ☐ B ☐ C ☐ D ☐

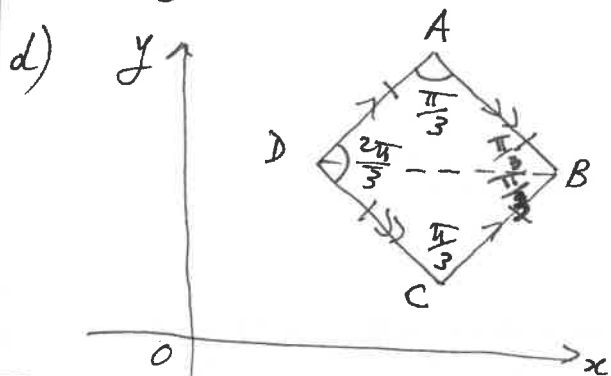
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Q11

$$\begin{aligned}
 \text{a) } \int \frac{x^3 dx}{x^4 + 3} &= \frac{1}{4} \int \frac{4x^3 dx}{(x^4)^2 + (\sqrt{3})^2} \\
 &= \frac{1}{4} \times \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^4}{\sqrt{3}} \right) + C \\
 &= \frac{1}{4\sqrt{3}} \tan^{-1} \left(\frac{x^4}{\sqrt{3}} \right) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int \cos^{-1} x dx & \quad \text{Let } u = \cos^{-1} x \\
 du &= \frac{-1}{\sqrt{1-x^2}} dx \\
 \int u dv &= uv - \int v du \quad dv = dx \\
 \therefore \int \cos^{-1} x dx &= x \cos^{-1} x - \int -x dx \\
 &= x \cos^{-1} x + \frac{1}{2} \int 2x (1-x^2)^{-\frac{1}{2}} dx \\
 &= x \cos^{-1} x - \frac{1}{2} (1-x^2)^{\frac{1}{2}} \times 2 + C \\
 &= x \cos^{-1} x - \sqrt{1-x^2} + C
 \end{aligned}$$

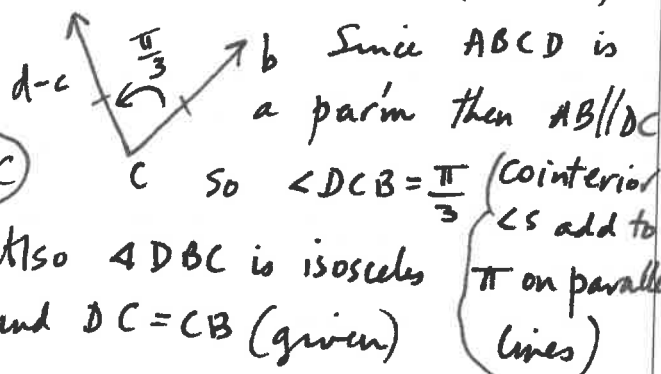
$$\begin{aligned}
 \text{c) } \int \sec^4 x dx &= \int \sec^2 x \sec^2 x dx \\
 &= \int (\tan^2 x + 1) \sec^2 x dx \\
 &= \int \tan^2 x \sec^2 x + \sec^2 x dx \\
 &= \frac{\tan^3 x}{3} + \tan x + C
 \end{aligned}$$



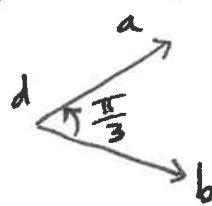
MC ANSWERS:

1. B 2. D 3. D 4. D 5. C
6. D 7. A 8. D 9. C 10. D

$$\text{d) i) RTP: } d-c = (b-c) \left(\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)$$



$$\begin{aligned}
 \text{ii) } \arg \left(\frac{a-d}{b-d} \right) &= \arg(a-d) - \arg(b-d) \\
 &= \frac{\pi}{3} \quad (\text{ABCD is a rhombus})
 \end{aligned}$$



$$\text{iii) } \left| \frac{b-a}{d-b} \right| = 1 \quad \text{since } \Delta ADB \text{ equilateral.}$$

$$\text{iv) } \arg \left(\frac{a-c}{b-d} \right) = \frac{\pi}{2} \quad \text{since ABCD is a rhombus. Diagonals bisect at right angles.}$$

Q12. a) $n \in \mathbb{N}$.

i) If n has exactly 2 factors then n is prime. (2)
($Q \Rightarrow P$)

ii) If n does not have exactly 2 factors then n is not prime.
($\neg Q \Rightarrow \neg P$) (2)

iii) Since $P \Rightarrow Q$ is true and $Q \Rightarrow P$ is true then $P \Leftrightarrow Q$ is true. It is an equivalence. (2)

iv) Statement not true. (2)

Counter-example is $n=1$ has 1 factor but is not composite.

b) At least one meerkat does not eat grubs. (2)

c) $x, y \in \mathbb{R}$:

$$x + 2y - 3xi + 4yi = 5 + 15i$$

Equate reals & imaginaries:

$$x + 2y = 5 \quad (1)$$

$$-3x + 4y = 15 \quad (2)$$

$$\begin{array}{r} (1) \times 2 \quad 2x + 4y = 10 \\ - (-3x + 4y = 15) \\ \hline \end{array}$$

$$5x = -5$$

$$x = -1$$

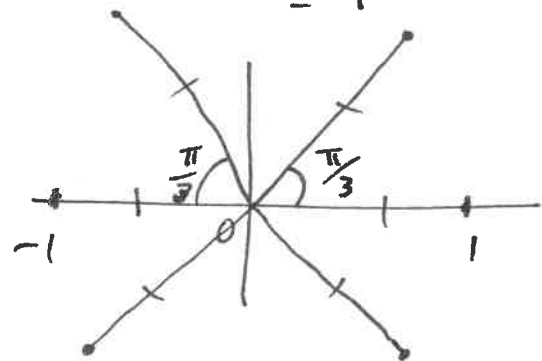
$$\Rightarrow (1) \quad -1 + 2y = 5 \quad (3)$$

$$2y = 6$$

$$y = 3.$$

$$\therefore x = -1, y = 3.$$

$$12 \text{ d) } (x + iy)^6 = e^{2i\pi} = 1$$



A solution could be

$$1 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = \frac{1}{2} + \frac{i\sqrt{3}}{2}$$

$$\therefore x = \frac{1}{2} \text{ and } y = \frac{\sqrt{3}}{2}.$$

(anything with mod 1 and $\arg \pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$) (roots equally spaced around 1 or -1)
(2)

Q13. a) $\underline{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$

$\underline{g} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

i) $\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$ AND

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$\therefore a = 1 + \lambda_1 \cdot 4 \quad \& \quad a = -1 + \lambda_2$

$b = -3 + \lambda_1(-5) \quad b = 6 + 2\lambda_2$

$c = 3 + \lambda_1(2) \quad c = 7 + 3\lambda_2$

$\therefore 1 + 4\lambda_1 = -1 + \lambda_2 \Rightarrow \lambda_2 = 2 + 4\lambda_1$

$\Rightarrow b = -3 - 5\lambda_1 = 6 + 2(2 + 4\lambda_1)$

$-3 - 5\lambda_1 = 6 + 4 + 8\lambda_1$

(2) $-13 = 13\lambda_1$

$\therefore \lambda_1 = -1, \lambda_2 = 2 + 4(-1) = -2$

$\therefore a = 1 + 4(-1) = -3$

$b = -3 - 5(-1) = 2 \quad \therefore \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}$

$c = 3 + 2(-1) = 1$

ii) RTP: $\underline{r} \cdot \underline{g} = 0$ or directions = 0

$\therefore 4 \times 1 - 5 \times 2 + 2 \times 3$

$= 4 - 10 + 6$

$= 0$

$\therefore$ Perpendicular. (2)

iii) let $\begin{pmatrix} l \\ m \\ n \end{pmatrix}$ be perp. to $\underline{r}$ & $\underline{g}$

$\therefore 4l - 5m + 2n = 0$ (1) and

$1l + 2m + 3n = 0$ (2)

Solve simultaneously:

(1) $\times 2 \quad 8l - 10m + 4n = 0$ (3)

(2) $\times 5 \quad 5l + 10m + 15n = 0$ (4)

(3) + (4) $13l + 19n = 0$

$\therefore l = \frac{-19n}{13}$

Let $n = 78$, then $l = -19 \times 6 = -114$

and $l + 2m + 3n = 0$

$-114 + 2m + 3 \times 78 = 0$

$2m = -120$ (3)

$m = -60$

$\therefore$ A vector $\perp$ $\underline{r} + \underline{g}$ is

$\begin{pmatrix} -114 \\ -60 \\ 78 \end{pmatrix}$ or $\begin{pmatrix} 19 \\ 10 \\ -13 \end{pmatrix}$

b) $P(5, 7, 2) \quad (x-1)^2 + (y+3)^2 + (z-3)^2 = k^2$

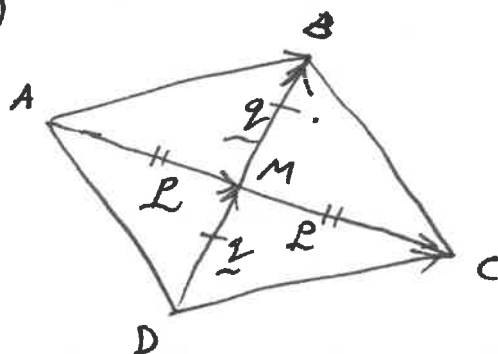
$\Rightarrow (5-1)^2 + (7+3)^2 + (2-3)^2 = k^2$

$4^2 + 10^2 + (-1)^2 = k^2$ (2)

$k^2 = 117$

$k = \sqrt{117} \quad (k > 0)$

c)



RTP: $\vec{AB} = \vec{DC}$

Proof: $\rightarrow$ over page:

Let $\vec{AM} = \underline{p} + \underline{MB} = \underline{q}$

$$\therefore \vec{MC} = \underline{p} \text{ and } \vec{DM} = \underline{q}$$

(given AC + BD bisected at M)

$$\text{Now in } \triangle AMB : \underline{p} + \underline{q} = \vec{AB}$$

$$\text{In } \triangle DMC, \underline{q} + \underline{p} = \vec{DC} \quad (3)$$

$$\therefore \vec{AB} = \vec{DC} \quad (\text{both } = \underline{p} + \underline{q}) \quad \text{QED.}$$

$$d) \text{ Let } \sqrt{15-8i} = a+ib, a, b \in \mathbb{R}$$

$$\therefore 15-8i = (a+ib)^2$$

$$\therefore 15-8i = a^2 + 2aib - b^2$$

$$\therefore 15-8i = a^2 - b^2 + 2aib$$

$$\text{Equating: } 15 = a^2 - b^2$$

$$-8 = 2ab \text{ or } -4 = ab$$

$$\text{By inspection, } a=4, b=-1$$

$$\text{or } a=-4, b=1$$

by convention, $a > 0$ so

$4-i$ is the root. (3)

Q14 a) i) RTP: $a^2 + b^2 \geq 2ab$ Proof: Consider the difference:

$$a^2 + b^2 - 2ab = (a-b)^2$$

$$\textcircled{1} \geq 0 \text{ (since square, equality when } a=b)$$

$$\therefore a^2 + b^2 \geq 2ab$$

ii) RTP: $a^2 + b^2 + c^2 \geq ab + bc + ca$.Proof: We know that

$$a^2 + b^2 \geq 2ab \quad \textcircled{2}$$

$$b^2 + c^2 \geq 2bc \text{ similarly,}$$

$$c^2 + a^2 \geq 2ca \quad \forall a, b, c \in \mathbb{R}$$

$$\text{Add: } 2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ca$$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

iii) RTP: $3(a^4 + b^4 + c^4) \geq (a^2 + b^2 + c^2)^2$.Proof: Consider the difference:

$$3(a^4 + b^4 + c^4) - (a^2 + b^2 + c^2)^2$$

$$\geq 3(a^2b^2 + b^2c^2 + c^2a^2) - (a^2 + b^2 + c^2)^2$$

from (ii)

$$\geq 3a^2b^2 + 3b^2c^2 + 3c^2a^2$$

$$- (a^2 + b^2 + c^2)(a^2 + b^2 + c^2)$$

$$\geq 3a^2b^2 + 3b^2c^2 + 3c^2a^2$$

$$- (a^4 + a^2b^2 + a^2c^2 + b^2a^2 + b^4 + b^2c^2$$

$$+ c^2a^2 + c^2b^2 + c^4)$$

$$\geq 3a^2b^2 + 3b^2c^2 + 3c^2a^2$$

$$- (a^4 + b^4 + c^4 + 2a^2b^2 + 2b^2c^2 + 2c^2a^2)$$

$$\geq a^2b^2 + b^2c^2 + c^2a^2 - a^4 - b^4 - c^4$$

$$= 3a^4 + 3b^4 + 3c^4 - (a^4 + b^4 + c^4 + 2a^2b^2 + 2b^2c^2 + 2c^2a^2)$$

14 a) iii) cont'd

$$= 2a^4 + 2b^4 + 2c^4 - (2a^2b^2 + 2b^2c^2 + 2c^2a^2)$$

$$\geq 2(a^2b^2 + b^2c^2 + c^2a^2) - 2(a^2b^2 + b^2c^2 + c^2a^2)$$

using (ii)

$$\geq 0 \text{ as required} \quad \textcircled{2}$$

$$\therefore 3(a^4 + b^4 + c^4) \geq (a^2 + b^2 + c^2)^2 \text{ QED!}$$

$$b) I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx, n \geq 0$$

$$i) \text{ RTP: } \frac{1}{n!} = e(I_{n-1} - I_n) \quad \textcircled{2}$$

Proof: $\int u dv = uv - \int v du$.

$$\text{Let } u = x^n \quad dv = e^{-x} dx$$

$$du = nx^{n-1} dx \quad v = -e^{-x}$$

$$\therefore \frac{1}{n!} \int_0^1 x^n e^{-x} dx = \frac{1}{n!} (uv - \int v du)$$

$$= \frac{1}{n!} \left[[x^n \cdot -e^{-x}]_0^1 - \int_0^1 -e^{-x} \cdot nx^{n-1} dx \right]$$

$$= \frac{1}{n!} \left[(1^n \cdot -e^{-1} - 0) + n \int_0^1 x^{n-1} e^{-x} dx \right]$$

$$= \frac{1}{n!} \left[-\frac{1}{e} + n \int_0^1 x^{n-1} e^{-x} dx \right]$$

$$\therefore I_n = -\frac{1}{en!} + \frac{1}{(n-1)!} \int_0^1 x^{n-1} e^{-x} dx$$

$$\therefore \frac{1}{en!} = I_{n-1} - I_n$$

$$\therefore \frac{1}{n!} = e(I_{n-1} - I_n) \text{ QED}$$

$$ii) I_0 = \frac{1}{0!} \int_0^1 x^0 e^{-x} dx$$

$$= [-e^{-x}]_0^1$$

$$= -\frac{1}{e} + 1$$

$$I_1 = I_0 - \frac{1}{e \cdot 1!} = 1 - \frac{1}{e} - \frac{1}{e} = 1 - \frac{2}{e}$$

$$I_2 = I_1 - \frac{1}{e \cdot 2!} = 1 - \frac{2}{e} - \frac{1}{2e} = 1 - \frac{5}{2e}$$

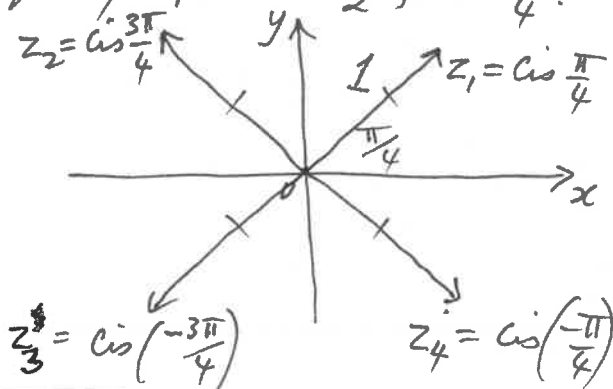
$$I_3 = I_2 - \frac{1}{e \cdot 3!} = 1 - \frac{5}{2e} - \frac{1}{6e} = 1 - \frac{16}{6e}$$

$$I_4 = I_3 - \frac{1}{e \cdot 4!} = 1 - \frac{16}{6e} - \frac{1}{24e} = 1 - \frac{65}{24e}$$

Q14 cont'd

c) Solve $z^4 = -1$.

i.e. $(\cos \theta + i \sin \theta)^4 = -1$ (3)

 $\therefore$ First $\theta = \frac{\pi}{4}$ thenequally spaced $\frac{\pi}{2}$ from $\frac{\pi}{4}$.d) RTP: $\sum_{k=1}^n (-1)^{k-1} k^3 = \frac{(2n-1)(n+1)^2}{4}$
ODD n.Proof: Let $P(n)$ be the proposition that

$$1^3 - 2^3 + 3^3 - 4^3 + \dots + n^3 = \frac{(2n-1)(n+1)^2}{4}$$

Prove $P(1)$ true:

$$\begin{aligned} \text{LHS} &= (-1)^{1-1} 1^3 \\ &= 1 \\ \text{RHS} &= \frac{(2(1)-1)(1+1)^2}{4} \\ &= \frac{(1)(2)^2}{4} \\ &= 1 \end{aligned}$$

 $\therefore P(1)$ true.Assume $P(k)$ true for some odd $k \in \mathbb{N}$:

$$1^3 - 2^3 + 3^3 - 4^3 + \dots + k^3 = \frac{(2(k)-1)(k+1)^2}{4}$$

RTP: $P(k+2)$ true i.e.:

$$\begin{aligned} 1^3 - 2^3 + 3^3 - 4^3 + \dots + k^3 - (k+1)^3 + (k+2)^3 &= \frac{(2(k+2)-1)(k+2+1)^2}{4} \\ &= \frac{(2k+3)(k+3)^2}{4} \end{aligned}$$

Proof: Consider the LHS of $P(k+2)$:

$$\begin{aligned} &1^3 - 2^3 + 3^3 - 4^3 + \dots + k^3 - (k+1)^3 + (k+2)^3 \\ &= \frac{(2k-1)(k+1)^2}{4} - (k+1)^3 + (k+2)^3 \\ &= \frac{(2k-1)(k+1)^2}{4} - \frac{4(k+1)^3}{4} + \frac{4(k+2)^3}{4} \\ &= (k+1)^2 \left[\frac{2k-1 - 4(k+1) + 4(k+2)^3}{4} \right] \\ &\quad + 4[k^3 + 2 \cdot 3k^2 + 2 \cdot 3k + 8] \\ &= (k+1)^2 \left[\frac{-2k-5}{4} \right] + \frac{4(k^3 + 6k^2 + 12k + 8)}{4} \\ &= \frac{-(k^2 + 2k + 1)(2k+5)}{4} + \frac{4(k^3 + 6k^2 + 12k + 8)}{4} \\ &= \frac{-(2k^3 + 5k^2 + 4k^2 + 10k + 2k + 5)}{4} \\ &\quad + \frac{4k^3 + 24k^2 + 48k + 32}{4} \\ &= \frac{2k^3 + 15k^2 + 36k + 27}{4} \\ &= \frac{(2k+3)(k+3)^2}{4} \quad \left[\text{Yes, it's true!} \right] \\ &= \text{RHS of } P(k+2). \end{aligned}$$

 $\therefore P(n)$ true by Math Induction.

(3)

Q15.a) i) $x + \frac{1}{x} = v$.

$$\left(x + \frac{1}{x}\right)^3 = x^3 + 3x^2 \cdot \frac{1}{x} + 3x \cdot \frac{1}{x^2} + \frac{1}{x^3}$$

$$= x^3 + \frac{1}{x^3} + 3x + \frac{3}{x}$$

$$\therefore x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3x - \frac{3}{x}$$

$$(2) = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= v^3 - 3v$$

ii) $x^5 + \frac{1}{x^5}$? Consider $\left(x + \frac{1}{x}\right)^5$

$$= x^5 + 5x^4 \cdot \frac{1}{x} + 10x^3 \cdot \frac{1}{x^2} + 10x^2 \cdot \frac{1}{x^3} + 5x \cdot \frac{1}{x^4} + \frac{1}{x^5}$$

$$= x^5 + 5x^3 + \frac{5}{x^3} + 10x + \frac{10}{x} + \frac{1}{x^5}$$

$$= x^5 + \frac{1}{x^5} + 5\left(x^3 + \frac{1}{x^3}\right) + 10\left(x + \frac{1}{x}\right)$$

$$\therefore x^5 + \frac{1}{x^5} = \left(x + \frac{1}{x}\right)^5 - 5\left(x^3 + \frac{1}{x^3}\right) - 10\left(x + \frac{1}{x}\right)$$

$$= v^5 - 5(v^3 - 3v) - 10v$$

$$= v^5 - 5v^3 + 15v - 10v$$

$$= v^5 - 5v^3 + 5v \quad (2)$$

iii) If $x = \cos \theta + i \sin \theta$ then

$$x + \frac{1}{x} = \cos \theta + i \sin \theta + \frac{1}{\cos \theta + i \sin \theta}$$

$$= \cos \theta + i \sin \theta + \cos \theta - i \sin \theta$$

$$\therefore v = 2 \cos \theta$$

$$\text{Now } x^5 + \frac{1}{x^5} = \cos 5\theta + i \sin 5\theta + \cos 5\theta - i \sin 5\theta$$

$$= 2 \cos 5\theta$$

$$\left(x^5 + \frac{1}{x^5}\right)^2 = x^{10} + 2 + \frac{1}{x^{10}}$$

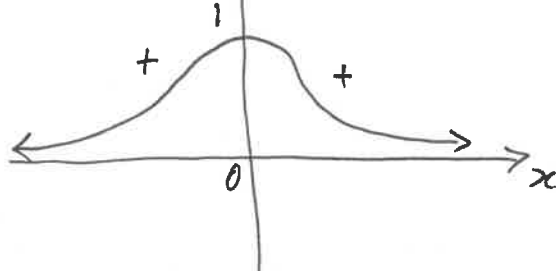
$$\text{So } x^{10} + \frac{1}{x^{10}} = \left(x^5 + \frac{1}{x^5}\right)^2 - 2$$

$$\text{but } \therefore 2 \cos 10\theta = (v^5 - 5v^3 + 5v)^2 - 2$$

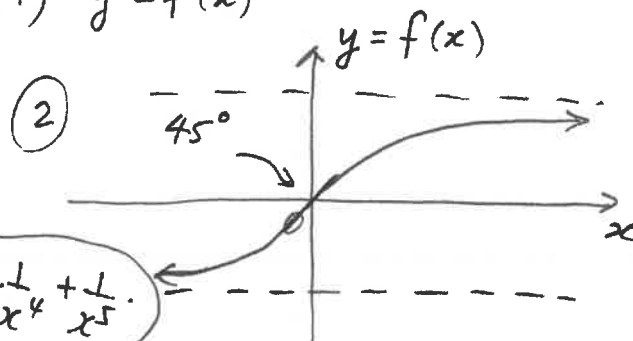
$$= \left((2 \cos \theta)^5 - 5(2 \cos \theta)^3 + 5(2 \cos \theta)\right)^2 - 2$$

$$\therefore \cos 10\theta = \frac{(32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta)^2 - 2}{2}$$

b) $y = f'(x)$



i) $y = f(x)$



ii) $g(x) = f^{-1}(x)$.

$$y = \frac{d}{dx} (g(x)) \quad (2)$$

$$\text{Now } \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$$

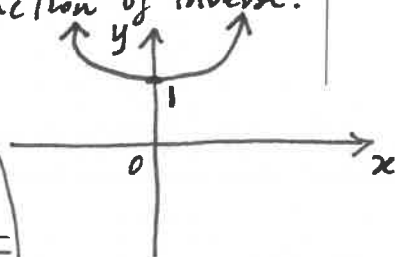
$x = g(y)$ is the same as

$x = f^{-1}(y)$ or $y = f(x)$

So $\frac{dx}{dy} = g'(y)$ Too Confusing using algebraic

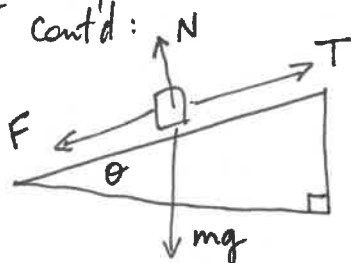
$\therefore \frac{dy}{dx} = \frac{1}{g'(y)}$ Proof unless use dummy variables.

Derivative $\frac{dy}{dx}$ is same as reciprocal function of inverse.



Q15 cont'd:

c)



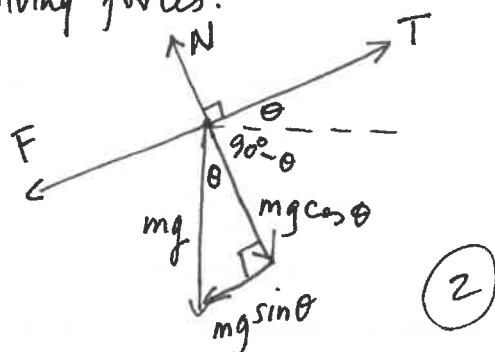
$$m = 2 \text{ kg}$$

$$F = \mu N$$

$$T = 10 \text{ N}$$

$$\mu = 0.3$$

Resolving forces:



Along the slope:

$$F_{\text{NET}} = T - F - mg \sin \theta$$

$$= 10 - \mu N - 2g \sin \theta$$

$$= 10 - 0.3N - 2g \sin \theta$$

Perpendicular to slope:

$$N = mg \cos \theta = 2g \cos \theta$$

$$\therefore F_{\text{NET}} = 10 - 0.3(2g \cos \theta) - 2g \sin \theta$$

$$= 10 - 2g(0.3 \cos \theta + \sin \theta)$$

d) cont'd:

$$u > 0$$

$$= \int_1^0 \frac{u}{\frac{1}{u} \sqrt{1+2u-u^2}} dx$$

$$= \int_1^0 \frac{u^2}{\sqrt{1-(u^2-2u)}} dx$$

$$= \int_1^0 \frac{u^2}{\sqrt{1-(u^2-2u+1-1)}} \times \frac{-1}{u} du$$

$$= \int_1^0 \frac{-du}{\sqrt{2-(u-1)^2}} \quad (3)$$

$$= \left[-\sin^{-1}\left(\frac{u-1}{\sqrt{2}}\right) \right]_1^0$$

$$= \left[\sin^{-1}\left(\frac{u-1}{\sqrt{2}}\right) \right]_0^1$$

$$= \sin^{-1}\left(\frac{1-1}{\sqrt{2}}\right) - \sin^{-1}\left(\frac{0-1}{\sqrt{2}}\right)$$

$$= 0 - -\frac{\pi}{4}$$

$$= \frac{\pi}{4}$$

$$d) u = \frac{1}{x} = x^{-1}$$

$$x = \infty$$

$$u = 0$$

$$du = -1x^{-2} = -\frac{1}{x^2} dx$$

$$x = 1$$

$$u = 1$$

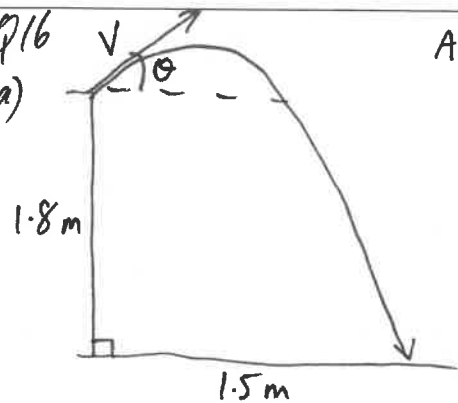
$$\int_1^{\infty} \frac{1}{x \sqrt{x^2+2x-1}} dx$$

$$dx = -x^2 du$$

$$= -\frac{1}{u^2} du$$

$$= \int_1^0 \frac{u}{\sqrt{\frac{1}{u^2} + \frac{2}{u} - 1}} dx$$

$$= \int_1^0 \frac{u}{\sqrt{\frac{1+2u-u^2}{u^2}}} dx$$

Q16
a)

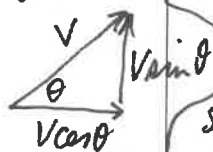
Assume

$\theta = 45^\circ$

$t = 0$

$x = 0$

$y = 1.8$



$$\ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$= C_1$$

$$\ddot{y} = -g$$

$$\dot{y} = \int -g dt \quad (4)$$

$$= -gt + C_2$$

When $t = 0$, $\dot{x} = V \cos \theta$, $\dot{y} = V \sin \theta$

So $\dot{x} = V \cos \theta = C_1$, $V \sin \theta = 0 + C_2$
 $C_2 = V \sin \theta$

$$\therefore \dot{x} = V \cos \theta \quad \dot{y} = -gt + V \sin \theta$$

$$x = \int V \cos \theta dt \quad y = \int -gt + V \sin \theta dt$$

$$= Vt \cos \theta + C_3 \quad y = -\frac{gt^2}{2} + Vt \sin \theta + C_4$$

When $t = 0$, $x = 0$, $y = 1.8$

$$\therefore 0 = 0 + C_3, \quad 1.8 = 0 + 0 + C_4$$

$$\therefore x = Vt \cos \theta, \quad y = -\frac{gt^2}{2} + Vt \sin \theta + 1.8$$

$$\therefore x = Vt \times \frac{1}{\sqrt{2}}, \quad y = -\frac{gt^2}{2} + Vt \times \frac{1}{\sqrt{2}} + 1.8$$

So $t = \frac{x\sqrt{2}}{V}$ and $x = 1.5, y = 0$

So $t = \frac{1.5\sqrt{2}}{V} \Rightarrow 0 = -\frac{g}{2} \left(\frac{1.5^2 \sqrt{2}}{V^2} \right) + \sqrt{\frac{1.5\sqrt{2}}{V^2}} + 1.8$

$$0 = -g \times \frac{9}{4V^2} + 3.3$$

$$\frac{9g}{4V^2} = \frac{33}{10} \Rightarrow \frac{4V^2}{9g} = \frac{10}{33}$$

$$\therefore V^2 = \frac{9 \times 9}{11 \times 33 \times 4} \quad \therefore |V| = \sqrt{\frac{159}{22}}$$

$$b) \tan^{-1} 4x - \tan^{-1} 3x = \tan^{-1} \frac{1}{7}$$

Using $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$:

Let $A = \tan^{-1} 4x$, $B = \tan^{-1} 3x$ then

$$\therefore \tan A = 4x, \quad \tan B = 3x$$

So $\tan(\tan^{-1} 4x - \tan^{-1} 3x)$
 $= \tan\left(\tan^{-1} \frac{1}{7}\right)$

$$\therefore \frac{4x - 3x}{1 + 4x \cdot 3x} = \frac{1}{7} \quad (\text{one-to-one})$$

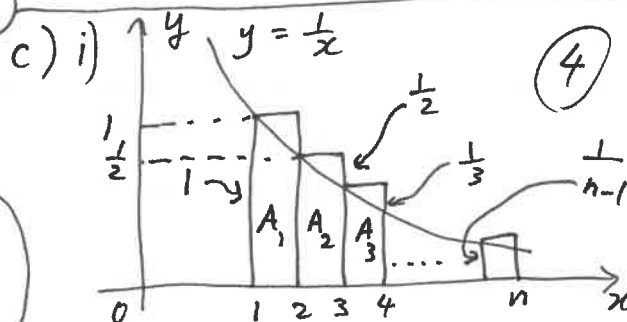
$$\frac{x}{1 + 12x^2} = \frac{1}{7} \quad (2)$$

$$\therefore 7x = 1 + 12x^2$$

$$12x^2 - 7x + 1 = 0$$

$$(4x - 1)(3x - 1) = 0$$

$$x = \frac{1}{4} \text{ or } \frac{1}{3}$$



RTP:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n-1} > \ln n$$

Proof: Sum of rectangles above curve $> \int_1^n \frac{1}{x} dx$

$$\therefore A_1 + A_2 + A_3 + \dots + A_{n-1} > [\ln x]_1^n$$

$$1 \times 1 + 1 \times \frac{1}{2} + 1 \times \frac{1}{3} + \dots + 1 \times \frac{1}{n-1} > \ln n - \ln 1$$

$$\therefore 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} > \ln n$$

Q16 cont'd

c) ii) Now $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} > \ln n$ So since $\ln n \rightarrow \infty$ as $n \rightarrow \infty$ and $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} > \ln n$ then $1 + \frac{1}{2} + \frac{1}{3} + \dots \rightarrow \infty$ as $n \rightarrow \infty$

So No limit. (1)

d) w satisfies $z^3 = 1$. $w \notin \mathbb{R}$.If $n \in \mathbb{N}$, then RTP: $1 + w^n + w^{2n} = 3$ if $n = 3X$ for
some $X \in \mathbb{N}$ OR $1 + w^n + w^{2n} = 0$ if $n \neq 3X$ ie. $n = 3X+1$ or $n = 3X+2$
in form.Proof: Consider $n = 3X$ so

$$1 + w^n + w^{2n} = 1 + w^{3X} + w^{2(3X)} \\ = 1 + (w^3)^X + (w^3)^{2X}$$

Since $w^3 = 1$ then

$$= 1 + 1^X + 1^{2X} \\ = 1 + 1 + 1 \\ = 3 \quad \checkmark \quad \text{QED}$$

Now Consider $n = 3X+1$ or $n = 3X+2$ So $1 + w^n + w^{2n} = 1 + w^{3X+1} + w^{2(3X+1)}$

$$= 1 + w^{3X} \cdot w^1 + w^{6X} \cdot w^2$$

$$= 1 + 1w + 1w^2$$

$$= 0 \quad \text{since } w^3 = 1$$

$$[(w^3 - 1)(w - 1)(w^2 + w + 1) = 0]$$

$$\text{OR } 1 + w^n + w^{2n} = 1 + w^{3X+2} + w^{2(3X+2)}$$

$$= 1 + w^{3X} \cdot w^2 + w^{6X} \cdot w^4$$

$$= 1 + 1 \cdot w^2 + 1 \cdot w^3 \cdot w$$

$$= 1 + w^2 + 1 \cdot w$$

$$= 1 + w + w^2$$

$$= 0 \quad \text{as well.}$$

$$\therefore 1 + w^n + w^{2n} = 3 \quad \text{if } n = 3X$$

$$\text{or } = 0 \quad \text{if not.}$$

QED.

Q14 (b) ii) better solution?

$$\frac{1}{n!} = e(I_{n-1} - I_n)$$

$$\frac{1}{1!} = e(I_0 - I_1)$$

$$\frac{1}{2!} = e(I_1 - I_2)$$

$$\frac{1}{3!} = e(I_2 - I_3)$$

$$\frac{1}{4!} = e(I_3 - I_4)$$

$$\therefore 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} = e \left(\begin{array}{l} I_0 - I_1 + I_1 - I_2 \\ + I_2 - I_3 + I_3 - I_4 \end{array} \right)$$

$$\frac{24+12+4+1}{24} = e(I_0 - I_4)$$

$$I_0 - \frac{41}{24e} = I_4$$

$$\therefore I_4 = 1 - \frac{1}{e} - \frac{41}{24e} \\ = 1 - \frac{65}{24e}$$

$$I_0 = \frac{1}{0!} \int_0^1 x^0 e^{-x} dx \\ = [-e^{-x}]_0^1 \\ = -e^{-1} + e^0 \\ = 1 - \frac{1}{e}$$

Student Number*SOLUTIONS*.....**ASCHAM SCHOOL****YEAR 12 Trial Mathematics Extension 2 Exam****MULTIPLE-CHOICE ANSWER SHEET**

1. A ☐ B ☒ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☐ B ☐ C ☐ D ☒
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☐ C ☒ D ☐
6. A ☐ B ☐ C ☐ D ☒
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☒
9. A ☐ B ☐ C ☒ D ☐
10. A ☐ B ☐ C ☐ D ☒